

The Evolution of Laryngeal Reinnervation, the Current State of Science and Thoughts for Future Treatments

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Summary: The treatment of unilateral vocal fold palsy (UVFP) or bilateral vocal fold palsy (BVFP) has been the subject of debate and experiment for 150 years. To date, dozens of different surgical methods have been described to re-innervate this most complex of organs, the larynx. As yet, there is no consensus on the most functionally effective method of reinnervation. However, it is a rapidly expanding area of research and remains an area of controversy. Indications for reinnervation for both UVFP and BVFP are still evolving and our understanding of the neuromuscular supply to the larynx continues to expand. What may have been considered unacceptable results from previous studies with one pathology may actually be of benefit in patients with different pathologies. This uncertainty of treatment options and potential outcomes can be confusing. In addition alternative techniques have been postulated as mainstays or adjuncts of treatment to the stalwart of reinnervation, neurotaphy. Determining what the correct treatment for an individual patient should be is still a gray area. With this in mind, this article reviews the evolution of laryngeal reinnervation, reviews the current state of the science, and suggests directions in which it might move in the future.

Key Words: Laryngeal reinnervation–Recurrent laryngeal nerve–Superior laryngeal nerve–Anastomosis of Galen–Neurotaphy–Vocal fold palsy.

INTRODUCTION AND ANTIQUITY

Galen (AD 129–199 [or 217]) is accredited with the first detailed description of the anatomy of the larynx and the description of laryngeal nerves. He understood that the voice came in part from vibration of the vocal folds. He dissected a series of mammals and discovered the same in each. Demonstrating on a pair of pigs to an audience in Rome, he described “... there is a pair [of nerves] in the muscles of the larynx on both left and right, which if ligated or cut render the animal speechless without damaging either its life or functional activity ...”¹ To further his demonstration of the nervous system, with the pig strapped to a table in a Roman piazza, he would take a knife and plunge it into the lower spine. The hind legs would stop moving but the forelegs and head continued to. He would then insert the knife into the lower neck through the cervical spine. The forelegs would now stop moving but the pig still breathed and cried out. Finally, he would insert the knife through the high cervical spine, killing the animal, and throwing his knife onto the table (Figure 1).

Although Galen accurately identified what are now named the recurrent laryngeal nerves (RLNs), there is no record of him describing the superior laryngeal nerves (SLNs).

The study of the larynx and its mechanisms of action, in comparison with the rest of the body, progressed slowly until direct visualization using light and mirror was first described by Bozzini in 1807. Unfortunately, colleagues condemned his invention of a speculum that reflected light and the idea was lost for almost 50 years but due to the tenacity and detailed presentations concerning the direct visualization of his own larynx, Garcia became known as the father of laryngoscopy and the technique gained a foothold.² The introduction of photographic

strobe lighting by Muybridge in the 1870s³ and its adaptation to visualize the larynx by Oertel in 1895⁴ allowed a greater understanding of the production of the voice. However, the clinician’s increasing interest in disorders of the larynx ran in parallel with these innovations and by the turn of the century, vocal fold palsy had been well described and attempts were being made to correct this defect.

ANIMAL EXPERIMENTS

In 1893, writing in *The Veterinarian*, Cotterell described the division of the left RLN of three dogs and a donkey and direct anastomosis with the divided ipsilateral vagus nerve. Two dogs died, one dog had a partially working, synkinetic left vocal fold (LVF) and the donkey had an LVF described as working synchronously with the right vocal fold (RVF).⁵ By any measure, this represents a partial success. One may consider that the sacrifice of one of the vagus nerves may have played a part in the demise of two of the dogs but with no published description, this is conjecture.

HUMAN CASE REPORT

The first description of a human laryngeal nerve reinnervation was by Horsley,⁶ of Richmond VA, in 1909 published in the *Annals of Surgery*. He described the presentation of a lady who had been shot by a pistol. The ball had entered:

... at the lower border of the chin about the median line and just grazing the bone. It was evidently deflected by the bone and took a course downward and to the left, just beneath the skin, to the larynx where it penetrated deeper in the neck. Just above the larynx the bullet so nearly penetrated to the surface that a keloid developed as a result of the injury to the deep layers of the skin. After striking the left side of the thyroid cartilage the bullet took a deeper course, wounding the left recurrent laryngeal nerve. There was only slight bleeding at the time, but the patient’s voice was at once affected and was so hoarse that she could not speak above a whisper.

The reason the woman had been shot was not mentioned. Horsley went on to operate in August 1908, excising the scarred

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FIGURE 1. Image from the bottom panel of the title page to the 1541 Junta edition of “Galen’s Works.” Depicts Galen demonstrating that the RLNs render an animal voiceless when cut.

portion of the RLN and anastomosing end-to-end with a sportingly large 0 catgut. Muscle was overlaid and the skin closed with silkworm gut. The patient was examined by Dr Clifton M. Miller, the regional Professor of Rhinology and Laryngology at the University of West Virginia after 2 and 15 months. At the first viewing, the LVF had slight movements, and at the second, there were almost normal movements. However, the LVF was noted to be lagging behind the RVF in its movements and there was an anteriorly based laryngeal web, presumably secondary to inflammation from the initial traumatic insult. One can assume that this was due to a mild degree of synkinesis.

To correct the lateralized vocal fold without reinnervating the affected muscles, surgeons started implanting material lateral to the fold in an attempt to metalize the fold into the midline. Bruening,⁷ in 1911, described the first injection medialization laryngoplasty using paraffin, and Payr,⁸ in 1915, the first external approach medialization thyroplasty. These techniques have remained the mainstay of treatment for unilateral vocal fold palsy (UVFP) until the present day.

ANATOMY

Every medical student knows that the RLN supplies motor innervation to all the muscle of the larynx except the cricothyroid and sensation to the larynx below the level of the vocal folds. The SLN supplies motor innervation to the cricothyroid and sensation larynx above the vocal folds. The folds themselves receive a mixed supply. However, as in most aspects of the human body, reality is not quite as simple as this. The RLN loops under the arch of the aorta on the left and under the subclavian artery on the right to run superiorly in the tracheo-oesophageal groove bilaterally. It curves antero-superiorly around the cricothyroid joint and enters the larynx. It divides, sometimes before and sometimes after the joint, into its adductor and abductor branches. The abductor branch runs posteriorly to supply the only abductor of the larynx, the posterior cricoarytenoid (PCA) and the adductor branch runs superiorly and further divides to supply the thyroarytenoid (TA) and interarytenoid muscles. Again, this division is not as simple in reality as it looks in the diagrams. There is cross-innervation of the PCA from the contralateral side and there

are often interdigitations between all these end branches. A suitable analogy would be the interconnecting branches of the facial nerve within the parotid gland, not described in textbooks, found during a parotidectomy.

As well as the complex subdivisions of the RLN, the SLN also plays a part. A well-described anastomosis between the two nerves is the appropriately named anastomosis of Galen. A recent review suggested an incidence of this connection of 81% in 50 larynges.⁹ In the event of RLN damage, the intrinsic muscles of the larynx will still receive some supply from the SLN. The review published in *Clinical Anatomy* also suggested that considering the morphometric contributions of the RLN and SLN, the SLN may be a larger contributor than previously thought. Which muscles this connection supplies and what ratio of abductor to adductor axons it carries are also unknown. Indeed, a 10-year study of the feline larynx published in 1992 by Yoshida et al¹⁰ focused on the myotopical arrangements of the motoneurons, an astonishingly intricate piece of work, but even so their summation was that “most of this innervation, ..., is unclear”. The fact that the SLN and the anastomosis of Galen may play a part intuitively feels right although there is no evidence either way at present.

HORSES

Laryngeal reinnervation in humans did not advance until the later part of the 20th century. Before this, a number of experiments on horses had been attempted because of an unusual inherited trait. Almost all Western thoroughbred horses are descended from three Arabian stallions imported over the last 300 years. The Byerley Turk (1680s), the Darley Arabian (1704), and the Godolphin Arabian (1729) were all stallions brought to the UK with an aim to sire winning thoroughbreds. With this small number of stallions, there has inevitably been inbreeding, a consequence of which is the incidence of clinical and subclinical left vocal fold palsy. It is postulated that the weakened LVF flaps in the airway when the horse is at gallop making a noise known as roaring. This limits the racing capacity of the horse in question and veterinarians have tried to circumvent this to push forward this sport of kings.

In 1936, Hobday¹¹ described a ventriculectomy performed through the cricothyroid membrane. During the late 20th century, a number of techniques were tried to address this problem including decompression, neurorrhaphy, neurolysis, and tracheostomy. Results were mixed. Arguably the best was tracheostomy simply bypassing the partial obstruction, but this decreased the respiratory dead space and gave these horses what was deemed by the racing authorities an unfair advantage. It has been banned in many countries and is being phased out by the British Horseracing Association as of October 2012.¹²

Reinnervation was first attempted using direct anastomosis, a nerve-muscle pedicle, and nerve implantation techniques in ponies.^{13–15} These attempts were partially successful but outcomes all describe a level of synkinesis present.

HARVEY TUCKER

The first concerted effort to answer the questions regarding laryngeal reinnervation was from Harvey Tucker from Cleveland, OH. He had been experimenting with reinnervation since 1970, and in 1976, he published a description of five patients with bilateral vocal fold palsy (BVFP) who underwent reinnervation of their PCA.¹⁶ The omohyoid branch of the ansa cervicalis was dissected free along with a small muscle pedicle. It was swung around posteriorly and sutured into the ipsilateral PCA. Abduction on inspiration was noted 6–8 weeks postoperatively. His review of 90 patients with BVFP treated with this same procedure in 1979 reported a 90% success rate.¹⁷

He also treated UVFP patients publishing an alternative target muscle using the same nerve/muscle pedicle technique.¹⁸ Instead of reinnervating the PCA with an aim to provide abduction of the vocal folds during inspiration in a patient with BVFP, the reinnervation was to be of the TA in those patients with UVFP, to facilitate movement of the fold toward the midline and regain voice and cough. It is interesting to note that same source, the omohyoid, was used for both abductor and adductor target muscles. Again, "... some degree of spontaneous adduction returned in all nine [of the patients] ..." and he reported good long-term outcomes using this technique in 25 out of 27 patients in 1981.¹⁹

These data suggested the possibility of a burgeoning area of laryngology. However, in other hands, this level of success was not reproducible. Something was still missing and this area of research was in danger of stalling. That was until the presentation of a Candidate's Thesis at the 85th Annual Meeting of the American Laryngological, Rhinological and Otological Society on May 5th 1982.²⁰

ROGER CRUMLEY

Roger Crumley had performed experiments on 26 dogs looking at different surgical techniques for the restoration of laryngeal function. He had carefully tested Dr Tucker's muscle pedicle technique of reinnervation but also tried, as a contrast, splitting the ipsilateral phrenic nerve and inserting it into the muscle belly of the PCA. The theory was that stimulation of the phrenic nerve to the diaphragm during inspiration would also stimulate the PCA causing abduction of the vocal folds—the larynx

would open. His results were fascinating; all the muscle pedicle subjects did not show any signs of abduction, contrary to the thinking of the time; however, the nerve insertion technique did. In those dogs that did not show any abduction it was shown, after the animal had been killed, that in all the cases, the nerve had become dislodged from the muscle. It was a tantalizing step forward.

SELECTIVE REINNERVATION

He moved his trials forward to human subjects and in 1983 published a case series of five patients with BVFP to whom he had performed a split phrenic nerve reinnervation to the PCA.²¹ He was circumspect in his reporting, stating "... no patient to date has demonstrated visible inspiratory vocal fold abduction however. The mechanism of action, if any, is unclear at this time and we have no electromyographic nor other physiologic data to confirm that true posterior cricoarytenoid muscle reinnervation has taken place ..." The effect on diaphragmatic function was not discussed.

In 1984, he changed tack and tried direct anastomosis of the nerves rather than implanting muscle. He attempted selective reinnervation in a canine model by continuing with his split phrenic nerve technique for the abductor mechanism and using nerve a graft from either the ansa or RLN or external branch of the SLN to the adductors.²²

Results showed that there was both adductor and abductor movements. However, in the discussion, it was postulated that the adductor reinnervation in some way inhibited the phrenic reinnervation of the abductor branch. However, in retrospect, it is more likely that some degree of synkinesis was present with cross-innervation between adductors and abductors.

Since the 1980s, Professor JP Marie, from Rouen, France has been grappling with the problem of selective reinnervation for BVFP. In both canine and leporine studies, during the 1980s and 1990s, his team persisted with selective reinnervation; in 1989, they published results from seven dogs with ansa—adductor branch and phrenic trunk—abductor branch. Adduction was present in five, abduction in only one. They stated, "... the reliability of the procedure can and must be improved ...".²³

A branch of the ansa cervicalis—either the descendens hypoglossi or the nerve to omohyoid—had been long thought of as the best nerve to produce adduction with acceptable morbidity from the nerve sacrifice and no studies since have contradicted this.

However, finding a suitable donor nerve to selectively reinnervate the abductor mechanism was more problematic. Previous studies had tried either the whole phrenic nerve or a split nerve technique. Although the results were not discussed in detail, it is apparent that taking the whole phrenic nerve would leave a significant respiratory morbidity for the patient and harvesting a split from the phrenic nerve would give the patient a variable and unpredictable insult to diaphragmatic movements. Professor JP Marie sought to discover the best and least injurious way to obtain an abductor nerve graft.

After a series of trials in rabbits during the 1990s with using different roots of the phrenic nerve, Professor JP Marie

published the results of a canine model that may give the right balance.²⁴ Using the most superior nerve root of the phrenic nerve and anastomosing it to the abductor mechanism of the larynx gave enough stimulation to the abductor muscles of the larynx without denervating the diaphragm. However, taking bilateral upper nerve roots gave a significant diaphragmatic functional defect. To avoid this, a unilateral nerve root would be taken and a cable graft used to innervate both PCA abductor muscles.

A human study showed that the phrenic nerve arises mainly from the C4 nerve root with variable C3 and C5 contributions.²⁵ Stimulation of each root showed that C4 was the main innervation to the diaphragm but that C5 can be of importance with stimulated pressure responses from the diaphragm almost equal to those produced from C4. C3 gave the least response. All the indications now suggested that the C3 nerve root would be the best for use in reinnervation of the abductor mechanism of the larynx. The question remained—what was the best anastomosis technique?

A review article by Marina et al²⁶ in 2011 also concluded that there was little data on reinnervation. However, Professor JP Marie's evolving technique was described. The results have yet to be published, but more than 20 patients have received ansa—adductor anastomoses and C3—bilateral abductor (using an ansa cable graft) implantations.

The technique was evolving but there was still the problem of synkinesis. Selective reinnervation still seemed to be the answer to BVFP, but synkinesis was proving to be a continuing problem. However, as is often the case when attempting to answer one question, an answer to an entirely different one was found as a by-product.

NONSELECTIVE REINNERVATION

The start of different way of using reinnervation arguably came about in 1986 when Professor Crumley published an article detailing the treatment of two patients he had treated having had high divisions of their vagus nerves after surgical treatment of vagal schwannomas.²⁷ Both were left with UVFP and both had a branch of the ansa cervicalis—RLN anastomosis. He reported "... although the reinnervated vocal fold neither abducted nor adducted, it presented itself in the midline for precise apposition with the nonparalyzed cord ..." In a follow-up report in 1991,²⁸ he described "... Excellent medialization of the paralyzed cord as well as correction of arytenoid malposition and thyroarytenoid muscle atrophy appear to explain the technique's success, since the reinnervated cord neither abducts nor adducts ..."

This is a classic description of synkinesis of the vocal fold and represented an alternative way of treating UVFP from the standard treatment of either injection (closed) or external approach (open) thyroplasty.

UVFP REINNERVATION

The interest in nonselective reinnervation continued and a number of case series with small numbers were published. Zheng et al²⁹ from Shanghai described ansa cervicalis—adductor

branch of RLN in eight patients with UVFP. They also found synkinesis although the anastomosis was with a terminal branch rather than the whole RLN. Position of the arytenoid was not discussed but acoustic analysis and stroboscopic analysis improved and the vocal fold retained tone and good mucosal waveform.

Lee et al³⁰ performed 25 ansa—RLN anastomoses for UVFP. Follow-up for 1 year revealed typical results from all these case series; improved voice outcomes in all patients. Acoustic and stroboscopic analysis improved; the patients were happy and breathiness improved 100% at 1 year.

Further studies have been published but the paucity of data was neatly encapsulated by a systematic review Aynehchi et al³¹ searching for reinnervation techniques for the management of UVFP. Only 14 articles fulfilled their search criteria encompassing 329 patients between 1966 and 2009. All were case series and the conclusions can be summarized as reinnervation may be effective for UVFP. No standard protocol for research or for the procedure is apparent and there are no good quality prospective studies. This is a problem that needs to be rectified and it is a situation that highlights the precarious position laryngeal reinnervation is currently at in its evolution.

SYNKINESIS

One of the reasons that a definitive technique for laryngeal reinnervation has yet to be achieved is the problem of synkinesis that has been alluded to throughout this article. Synkinesis is the involuntary movements of muscles that accompany voluntary movements. The RLN supplies both adductor and abductor muscles of the larynx with an axonal distribution of 3:1 within the neuron. The distribution of these axons is random and so determining which part of the nerve supplies which muscle is currently not possible. With a neuropraxic nerve injury (damage to the axonal tissue without disruption of the surrounding myelin sheath), the neuronal tissue attempts regeneration along the axonal sheath. However, most regenerating fibers are unable to penetrate the scar. Those that do get through are likely to innervate the wrong motor unit so there will be a misconnection between the impulses generated in the central nervous system and the muscles that are stimulated. This means that both adductors and abductors contract against each other producing uncoordinated movement—synkinesis. In addition, there may be neuroma formation, an inevitable loss of fiber population and the aforementioned inappropriate reinnervation.

If one is aiming for perfect movements of vocal folds, synkinesis is something to be avoided. However, in the case of a UVFP synkinesis may actually benefit the patient. For an explanation, one must look the current treatments used for patients with both UVFP and BVFP.

In the case of BVFP, the constant debate is between maintaining an adequate airway and regaining reasonable voice while keeping competency of the larynx (an ability to cough and clear secretions). A typical presentation would be of a patient with BVFP in respiratory distress, needing a tracheostomy to maintain an airway. Once stabilized, the patient could choose between either a permanent tracheostomy (maintaining both

airway and voice) or an arytenoidectomy to improve the airway at the glottis (with an aim to remove the tracheostomy) but with some reduction in voice. To reinnervate the larynx in BVFP, the aim is to have symmetrically moving vocal folds that both adduct and abduct, both maintaining airway and voice.

In the case of UVFP, the airway is seldom a concern as one vocal fold is moving normally. Therefore, the aim of any treatment is to medialize the vocal fold to allow the contralateral fold to oppose it, regaining voice. Traditional surgical treatment includes either injection (closed) or implantation (open) thyroplasty. Nonselective reinnervation of the larynx, with its attendant synkinesis, has already been shown to achieve this goal. Reinnervation would take approximately the same length of surgical time as an external approach thyroplasty would cost less as there is no implant material involved and, therefore, the risks of implant extrusion, infection, and perforation are all eliminated. However, the actual nerve regrowth takes time and final results might take 9 months to fully appreciate. Would this be time the patient is willing to wait? Is nonselective reinnervation for UVFP now a viable alternative to traditional procedures for patients?

AIMS FOR CURRENT AND FUTURE RESEARCH

The concept of reinnervating the larynx is an old one and intuitively feels right. Rather than implant foreign material into the body, can we reanimate normal mechanisms? Currently, the technique is not perfect but is under constant evolution. The author suggests that nonselective reinnervation may be considered a viable alternative to traditional surgical treatment for UVFP as long as the patient is prepared to wait for results. The problem of synkinesis, in this case, is an advantage over traditional methods and with little to no cost implication. However, selective reinnervation remains the goal for BVFP. The elaborate nature of the laryngeal neuroanatomy with its many interdigitations and anastomoses gives a complex problem to the surgeon and the perfect solution has yet to be discovered.

Therefore, future research will be in two main areas: further defining the best selective reinnervation technique for BVFP and attempting to speed up the nerve regrowth once surgical anastomosis has taken place. The first I have discussed and the second is an area that stems from both central nervous system and transplant medicine. An article published by Gorphe et al.³² in 2012 has shown that the immunosuppressive drug tacrolimus may aid reinnervation in rats. The tacrolimus group had significantly more normal laryngeal muscle architecture at 45 days than groups using low-dose tacrolimus, low-dose cyclosporin or surgery. The ethics of giving an immunosuppressant to an otherwise well patient are to be debated; however, anatomically localized drug administration may also play a part. Drug eluting stents are used in vascular and cardiac pathology and similar concepts might be used in the future to create drug-eluting channels through which nerves may grow.

SUMMARY

Whatever the research will show, it remains the fact that reinnervation of the larynx is a surgical concept that is here to

stay. Nonselective reinnervation of the larynx may be considered as an alternative to traditional surgical treatment for UVFP. Selective reinnervation of the larynx for BVFP remains the goal for restoring normal function of the larynx and promising results suggest that the future of neurolaryngeal surgery is bright.

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